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(Started from 8th October 2020)

- ✓✓✓ 1. **Understanding computer programs** (First week - 8/10/2020 - 14/10/2020)
 - ✓ - What is computer program and algorithm
 - ✓ - What is flowchart
 - ✓ - What is pseudocode
 - ✓ - Examples
 - ✓ - Setting up c++ environment
 - ✓ - Hello world program
 - ✓ - Brain teasers
- ✓ 2. **Programming Fundamentals I** (First week)
 - Data Types and ranges
 - Type modifiers
 - Input output in C++
 - If else
 - Loops - while, do while, for
- ✓ 3. **Programming Fundamentals II** (First week)
 - Break and continue
 - Switch
 - Operators - arithmetic, relational, logical, bitwise, assignment
- ✓ 4. **Pattern Questions** (Second week 15/10/2020 - 21/10/2020)
 - Introductory examples
 - Difficult examples
- ✓ 5. **Basics of Number System** (Second week)
 - Binary number system
 - Reverse a number
 - Armstrong number
 - Prime numbers
- ✓ 6. **Functions** (second week) (Second week)
 - Introduction to functions
 - Examples
 - ~~conversions~~
- ✓ 7. **Space and time complexity** (Third week 22/10/2020 - 28/10/2020)
 - Definition and basics
 - Polynomial evaluation
- ✓ 8. **Arrays** (Third week)

- ✓ Introduction to arrays
- ✓ Linear search and Binary search
- ✓ Sorting algorithms
- ✓ STL sort
- ✓ Max subarray sum
- ✓ Pair sum problem

9. 2D arrays (Third week)

- Introduction and searching
- Example questions

10. Character arrays (Fourth week 29/10/2020 - 4/11/2020)

- ✓ Introduction and input
- ✓ Palindromic arrays
- ✓ Largest word in a string

11. Strings (Fourth week)

- Intro + STL functions on strings
- Sorting a string

12. Questions (Fourth week)

- Target sum triplets
(<https://www.geeksforgeeks.org/find-a-triplet-that-sum-to-a-given-value/>)
- ✓ Max circular sum
(<https://www.geeksforgeeks.org/maximum-contiguous-circular-sum/>)
- String questions

13. Pointers (Fourth week)

- ✓ & operator
- ✓ Introduction to pointers
- ✓ Dereference operator
- ✓ Pass by reference and Pass by value

14. Dynamic Memory Allocation (Fourth week)

- ✓ Compile time and run time
- ✓ Heap and Stack
- ✓ New, delete

15. Bit Manipulation (Fifth week 5/11/2020 - 11/11/2020)

- ✓ Get, set, clear, update
- ✓ Is Power of 2
- ✓ Number of ones
- ✓ Generate subsets

✓ 16. Questions (Fifth week)

- ✓ Unique number in an array of duplicates
- ✓ 2 unique numbers in an array of duplicates
- ✓ Unique number in an array of triplets

✓ 17. Prime Sieve (Fifth week)

- Prime sieve

✓ 18. Number Theory basics (Fifth week)

- ✓ Euclid's algo for GCD
- ✓ Inclusion exclusion principle

✓ 19. Introduction to Recursion (Fifth week)

- ✓ Call stack
- ✓ Fibonacci numbers
- ✓ Factorial
- ✓ Fast power
- ✓ First occurrence and last occurrence
- ✓ Increasing, Decreasing order
- ✓ Time complexity for recursive functions (Master's theorem)

✓ 20. Recursion - II (Fifth week)

- ✓ Tower of hanoi
- ✓ Reverse string
- ✓ Replace pi
- ✓ Remove duplicates
- ✓ Move all x
- ✓ Subsequence generation
- ✓ Generate permutations

✓ 21. Recursion - III (Fifth week)

- ✓ Permutation
- ✓ Board game
- ✓ 0-1 Knapsack
- ✓ Tiling problem
- ✓ Friends pairing problem
- ✓ Count paths in Maze

22. Backtracking (Sixth week 12/11/2020 - 18/11/2020)

- Rat in maze problem
- N-queen problem

✓ 23. Divide and Conquer (Sixth week)

- ✓ Merge sort
- ✓ Quick sort

✓ 24. More sorting techniques (Sixth week)

- ✓ Counting sort
- ✓ DNF algorithm
- ✓ Wave sort

✓ 25. OOPS concepts (Sixth week)

- ✓ Classes and objects
- ✓ Data members and functions
- ✓ Getters, setters.
- ✓ Constructor & its types
- ✓ Shallow and deep copy
- ✓ Copy assignment
- ✓ Destructor
- ✓ Overloading

26. STL (Sixth week)

- ✓ Pair class
- Other STL functions, Iterators, comparators 3
- Templates

✓ 27. Vectors (Sixth week)

- ✓ Intro
- ✓ Methods
- ✓ Sorting
- ✓ Template

✓ 28. Linked List (Sixth week)

- ✓ Introduction and implementation
- ✓ Insertion in linked list
- ✓ Searching in linked list
- ✓ Deletion in linked list
- ✓ Reverse a linked list - Iterative and recursive solution
- ✓ K reverse problem
- ✓ Floyd's cycle detection and removal
- ✓ Doubly linked list
- Circular linked list

29. Challenges (Seventh week 19/11/2020 - 25/11/2020)

- K append

- Even after odd
- Intersection point
- Merge 2 sorted linked lists

30. Stacks (Seventh week)

- Introduction
- Implement using array
- Implement using linked list
- Reverse a stack
- Balanced parenthesis

31. Infix, prefix, postfix (Seventh week)

- Infix, prefix, postfix expression evaluation
- Infix to prefix using stack
- Infix to postfix using stack

32. Queues (Seventh week)

- Introduction
- Implement using array
- Implement using linked list
- Implement stack using queue
- Implement queue using stacks

33. Deque (Seventh week)

- Introduction
- Maximum element
- Max length unique character substring

34. Questions (Eighth week 26/11/2020 - 02/12/2020)

- Histogram area
- Circular tour
- Balanced parentheses

35. Binary Trees (Eighth week)

- Introduction
- Preorder, inorder, postorder
- Level order
- Sum at level K
- Height and Diameter of Binary Tree
- BFS traversal
- DFS traversal
- Count and sum nodes
- Height balanced tree
- Build balanced tree from array

- Different views of binary tree
- Nodes at distance K
- Lowest common ancestor

36. Questions (Eighth week)

- Build from inorder and preorder
- Sum at level K
- Sum replacement problem
- Maximum sum path
- Shortest distance between nodes

37. Binary Search Tree (Ninth week 03/12/2020 - 09/12/2020)

- Introduction
- Implementation and insertion
- Searching
- Deletion
- Check for BST
- Find min and max element
- Flatten a tree
- Construct from preorder
- Catalan no concept
- Set STL

38. Questions (Ninth week)

- Structurally identical BST
- ZigZag order
- Largest BST in BT

39. Heaps (ninth week)

- Introduction to priority queue
- Heaps, insertion
- Remove min and max element
- Build heap from array
- Heapsort
- Priority queue STL, Running median

40. Challenges (ninth week)

- Top k most frequent numbers in stream
- Merge k sorted arrays
- Length of Smallest Subsequence such that sum of elements is greater than equal to K

41. Hashtable (Tenth week 10/12/2020 - 16/12/2020)

- Introduction to hash functions

- Collision handling and separate chaining
- Rehashing, load factor
- Unordered Map STL
- Max frequency character
- Vertical order print

42. Hashing Problems (Tenth week)

- Number of subarrays with sum 0
- Longest subarray with sum k
- Longest consecutive subsequence
(<https://www.geeksforgeeks.org/longest-consecutive-subsequence/>)
- Minimum window substring

43. Greedy Algorithm (Tenth week)

- Introduction
- Activity selection problem
- Job selection problem
- 0/n knapsack problem
- Optimal merge pattern problem
- Huffman coding problem

44. Challenges (Tenth week)

- Coin change
- Max Circles
- Dividing array

45. Dynamic Programming (Eleventh week 17/12/2020 - 23/12/2020)

- Introduction
- Fibonacci problem
- Minimum steps to 1
- Minimum coin change
- Maximum subarray sum
- Snakes and Ladders
- 0/1 knapsack
- LIS and LCS problem
- Matrix chain multiplication
- Friends pairing problem
- Catalan number concept
- Optimal game strategy
- Optimal binary search tree
- All pair shortest path problem

46. Challenges (Eleventh week)

- No. of Binary String
- LCS w 3 strings
- Wildcard pattern matching

- Brackets all over
- Max length bitonic subsequence
- Max sum submatrix

(<https://www.geeksforgeeks.org/maximum-sum-rectangle-in-a-2d-matrix-dp-27/>)

47. Graphs - I (Twelfth week 24/12/2020 - 30/12/2020)

- Introduction
- Representation
- Adjacency list implementation
- BFS
- DFS
- Topological sort
- Cycle detection in directed and undirected graph
- Connected components
- Pairing problem
- Bipartite graph check

48. Graphs - II (Twelfth week)

- Disjoint set introduction
- Union and find
- Path compression
- Union by rank optimisation
- Implementation

49. Graphs - III (Twelfth week)

- Kruskal's algorithm
- Prim's algorithm
- Dijkstra's algorithm
- Bellman ford algorithm
- Floyd Warshall algorithm
- Strongly connected component using Kosaraju's algo

50. Challenges (Twelfth week)

- Snakes and ladders problem
- MST problem
- Beautiful vertices

51. Questions (Thirteenth week 31/12/2020 - 6/01/2021)

- Sum of all submatrices in a matrix
- Searching in sorted matrix
- Rain water harvesting (<https://www.geeksforgeeks.org/trapping-rain-water/>)

52. Number theory Advanced (Thirteenth week)

- Extended Euclid
- Multiplicative modulo inverse
- Euler totient function
- Segmented sieve

- Binary/Modular exponentiation - both recursive and iterative
- Matrix Exponentiation - It's cases
- Fermat little theorem, wilson theorem

53. Tries (Thirteenth week)

- Data structure introduction
- Insertion
- Searching
- Phonebook problem
- Xor subarray problem

54. Questions (Thirteenth week)

- Intersection of 2 arrays
- String window
- Subarrays with distinct element
- Digital dictionary

55. String algorithms (Thirteenth week)

- Brute force
- KMP
- Finite automata
- Robin karp

56. Segment tree (Fourteenth week 7/01/2021 - 13/01/2021)

- Intro, build, updation, query
- Min, max, sum in the subarray
- Lazy propagation

57. Binary indexed tree (Fourteenth week)

- Structure of BIT
- Update or build, query.
- Problems

58. Mo's (sqrt decomposition) (Fifteenth week 14/01/2021 - 20/01/2021)

- Offline processing of queries with examples
- Introduction, code and examples

59. HLD (Fifteenth week)

60. Fourier series (Fifteenth week)